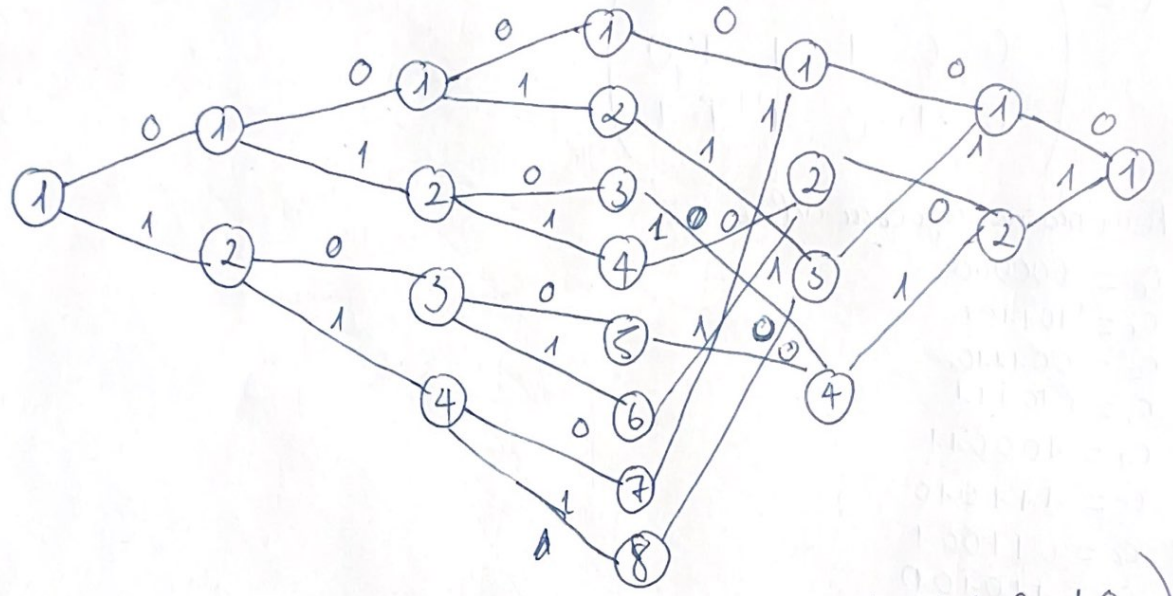
$$G = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix}$$

1. Решётка по кодовым словам

$c_0 = 000000$
$c_1 = 101101$
$c_2 = 001110$
$c_3 = 010111$
$c_4 = 100011$
$c_5 = 111010$
$c_6 = 011001$
$c_7 = 110100$

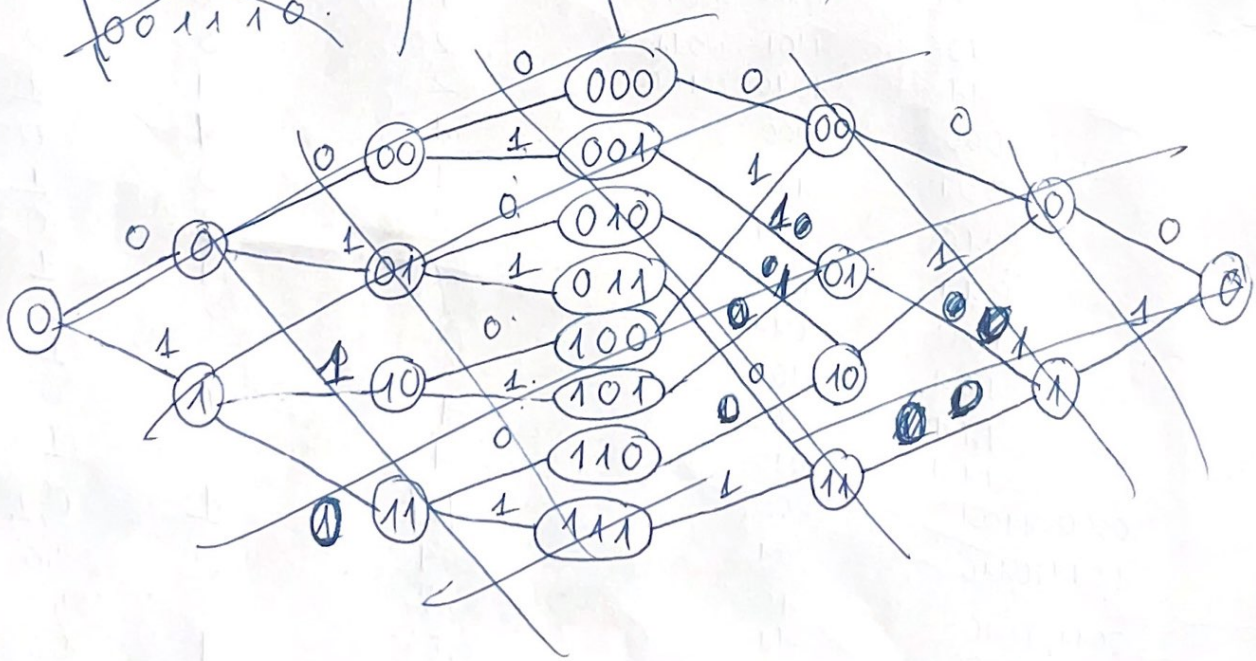
| $i$ | $\{c^p\}$ | $F_i(c_i^p)$ | $\{v_{i-1}\}$ | $\{v_i\}$ | $c_i$ |
|---|---|---|---|---|---|
| 0 | $\{\emptyset\}$ | $\{c_m\}$ | $\emptyset$ | 1 | — |
| 1 | 0 | 00000 01110 | 1 | 1 | 0 |
| | | 10111 11001 | | | |
| | 1 | 01101 00011 | 01 | 2 | 1 |
| | | 11010 10100 | | | |
| 2 | 00 | 0000 1110 | 1 | 1 | 0 |
| | 01 | 0111 1001 | 1 | 2 | 1 |
| | 10 | 1101 0011 | 2 | 3 | 0 |
| | 11 | 1010 0100 | 2 | 4 | 1 |
| 3 | 000 | 000 | 1 | 1 | 0 |
| | 001 | 110 | 1 | 2 | 1 |
| | 010 | 111 | 2 | 3 | 0 |
| | 011 | 001 | 2 | 4 | 1 |
| | 100 | 011 | 3 | 5 | 0 |
| | 101 | 101 | 3 | 6 | 1 |
| | 110 | 100 | 4 | 7 | 0 |
| | 111 | 010 | 4 | 8 | 1 |
| 4 | 0000, 1101 | 00 | 7,7 | 1 | 0,1 |
| | 1011, 0110 | 01 | 6,4 | 2 | 1,0 |
| | | 10 | 2,8 | 3 | 1,0 |
| | 0011, 1110 | 11 | 3,5 | 4 | 1,0 |
| | 0101, 1000 | | | | |
| 5 | 00000, 11010, 00111, 11101 | 0 | 10,3 | 1 | 0,1 |
| | 10110, 01100, 01011, 10001 | 1 | 2,4 | 2 | 0,1 |

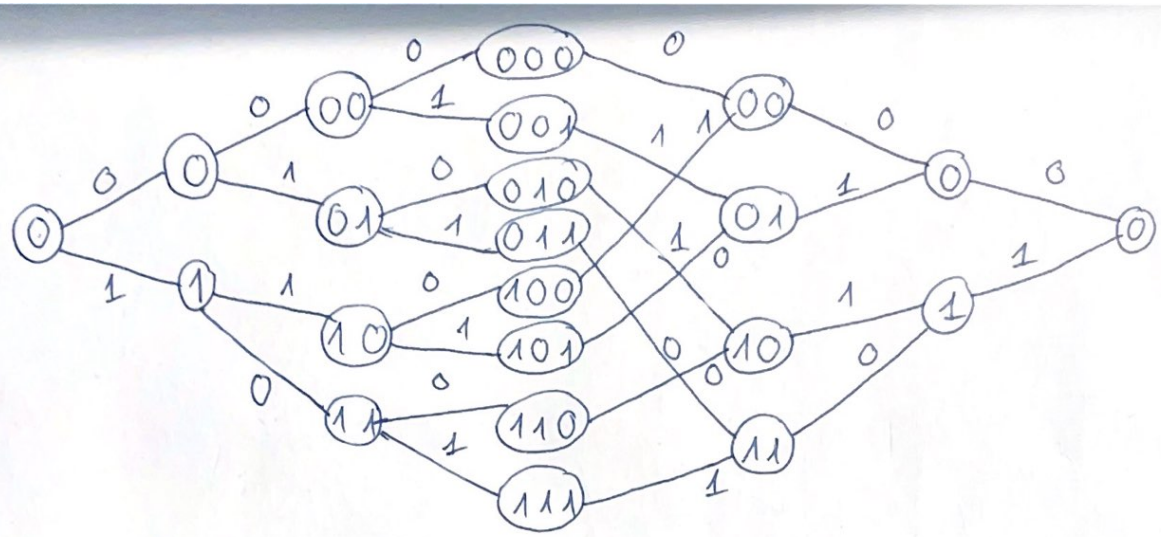
6 {cm} {ϕ} 1,2 1 0,1



$$G = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix} \overset{g_2 \leftrightarrow g_3}{\sim} \begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix} \overset{g_1 \leftarrow g_1 + g_2}{\sim} \begin{pmatrix} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix}$$

$$\overset{g_1 \leftarrow g_1 + g_3}{\sim} \cancel{\begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix}} \qquad \overset{g_1 \leftarrow g_1 + g_3}{\sim} \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix}$$

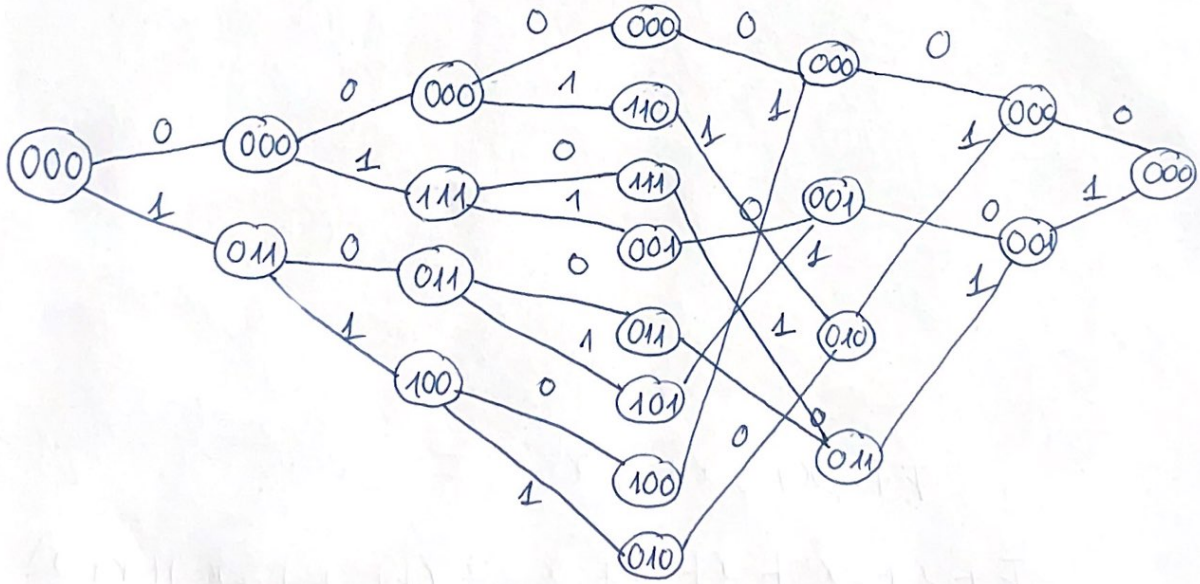





$$H = \begin{pmatrix} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 000 | 000 | 000 | 000 | 000 | 000 |
| | 011 | 111 | 001 | 001 | 001 |
| | | 011 | 010 | 010 | |
| | | 100 | 011 | 011 | |
| | | | 100 | | |
| | | | 101 | | |
| | | | 110 | | |
| | | | 111 | | |

6. 000.




Видим, что решетки совпадают с точностью до нумерации узлов.